

LOW ACRYLONITRILE NBR PATENT LANDSCAPE REPORT

1. ABSTRACT

This report reviews the provided patent evidence for low acrylonitrile nitrile butadiene rubber (NBR) synthesis and processing, with emphasis on how acrylonitrile content is controlled to balance low-temperature flexibility and oil resistance. The single patent in the manifest, EP2145681B1, is not a direct low-ACN copolymerization disclosure; instead, it addresses post-polymerization molecular-weight reduction of nitrile rubber by metathesis using ruthenium- or osmium-based catalyst systems combined with Lewis-acid/acid adducts. The patent defines NBR broadly as copolymers or terpolymers of at least one α,β -unsaturated nitrile and at least one conjugated diene, and it is relevant because lower molecular weight NBR/HNBR grades are often sought for improved processability in low-ACN elastomer applications. The disclosed process is aimed at reducing Mooney viscosity and number-average molecular weight of nitrile rubber feedstocks prior to or in conjunction with hydrogenation, rather than changing acrylonitrile feed ratio during polymerization. The evidence highlights the use of metathesis catalysts based on ruthenium or osmium, with optional compounds of formula (Z) such as BF_3 , BF_2Cl , BFCl_2 , BF_2Br , BFBr_2 , $\text{BF}_2(\text{OC}_2\text{H}_5)$, $\text{BF}(\text{OC}_2\text{H}_5)_2$, $\text{BF}_2(\text{CH}_3)$, or $\text{BF}(\text{CH}_3)_2$, combined with donor solvents or acids including water, diethyl ether, ethylamine, THF, n-propanol, formic acid, acetic acid, trifluoroacetic acid, trichloroacetic acid, sulfuric acid, phosphoric acid, trifluoromethanesulphonic acid, or toluenesulphonic acid. The broader technical context notes that commercially available HNBR often has Mooney viscosity ML 1+4 at 100°C of 55 to 120 and M_n of about 200,000 to 700,000, while lower-viscosity materials below ML 55 or M_n below 200,000 are desirable for processing. Overall, the patent landscape here supports downstream molecular-weight tailoring of nitrile rubbers, but does not provide direct low-ACN copolymerization recipes.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: EP2145681B1
- Patent Title: Method for the reduction of the molecular weight of nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2013

- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses a process for reducing the molecular weight of nitrile rubber by contacting the rubber with a catalyst system and reacting it by metathesis. The catalyst system includes a metathesis catalyst, typically based on ruthenium or osmium, together with at least one compound of formula (Z) such as a BF₃-derived adduct or related Lewis-acid/acid combination. - Hydrogenation degree of diene units: 50 to 100% — background description of HNBR - Mooney viscosity (ML 1+4 at 100°C): 55 to 120 — background description of commercial HNBR - Number-average molecular weight Mn: about 200,000 to 700,000 — background description of commercial HNBR - Residual double bond content: 0 to 18% — background description of commercial HNBR - Mooney viscosity target: less than 55 — low molecular weight HNBR discussion - Number-average molecular weight target: Mn < 200,000 g/mol — low molecular weight HNBR discussion - Weight-average molecular weight Mw: 30,000 to 250,000 — cited prior-art low-Mooney HNBR process - Mooney viscosity (ML 1+4 at 100°C): 3 to 50 — cited prior-art low-Mooney HNBR process - Polydispersity index PDI: less than 2.5 — cited prior-art low-Mooney HNBR process - Grubbs-I catalyst loading: 0.5 phr — cited prior-art discussion - Ruthenium content: 61 ppm Ru — cited prior-art discussion

Relevant Experimental Evidence - Example block: no discrete numbered examples were detected by the parser in the provided evidence. - Background: the patent explains that nitrile rubber is a copolymer or terpolymer of an α,β -unsaturated nitrile and a conjugated diene. - Background: the patent states that lower molecular weight nitrile rubber is desirable because it improves processability. - Background: prior-art metathesis/hydrogenation routes are discussed as a route to low-Mooney HNBR, including specific molecular-weight and viscosity ranges.

Technical Relevance This patent is relevant because it provides a route to lower molecular weight nitrile rubber, which is directly tied to processability in low-ACN NBR/HNBR systems. Although it does not change acrylonitrile content during polymerization, it supports downstream tailoring of nitrile rubber properties for low-ACN applications.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- No qualifying primary patents were identified under the configured target material relevance criteria. Cross-patent quantitative trends were therefore not generated.

4. CONCLUSION

The evidence set contains one patent focused on post-polymerization molecular-weight reduction rather than direct low-acrylonitrile copolymer synthesis. EP2145681B1 is technically useful for

low-ACN NBR development because it addresses the processing bottleneck of high molecular weight nitrile rubber and provides catalyst-system concepts for lowering Mooney viscosity and Mn. Based on the evidence, the most defensible synthesis/process recommendation is to use nitrile rubber feedstocks with controlled composition, then apply metathesis-based molecular-weight reduction when lower viscosity is required; however, no direct acrylonitrile feed-ratio recipe is disclosed in the provided patent.

5. REFERENCES

- EP2145681B1 | Method for the reduction of the molecular weight of nitrile rubber | Not disclosed | EP | 2013 | <https://patents.google.com/patent/EP2145681B1/en>